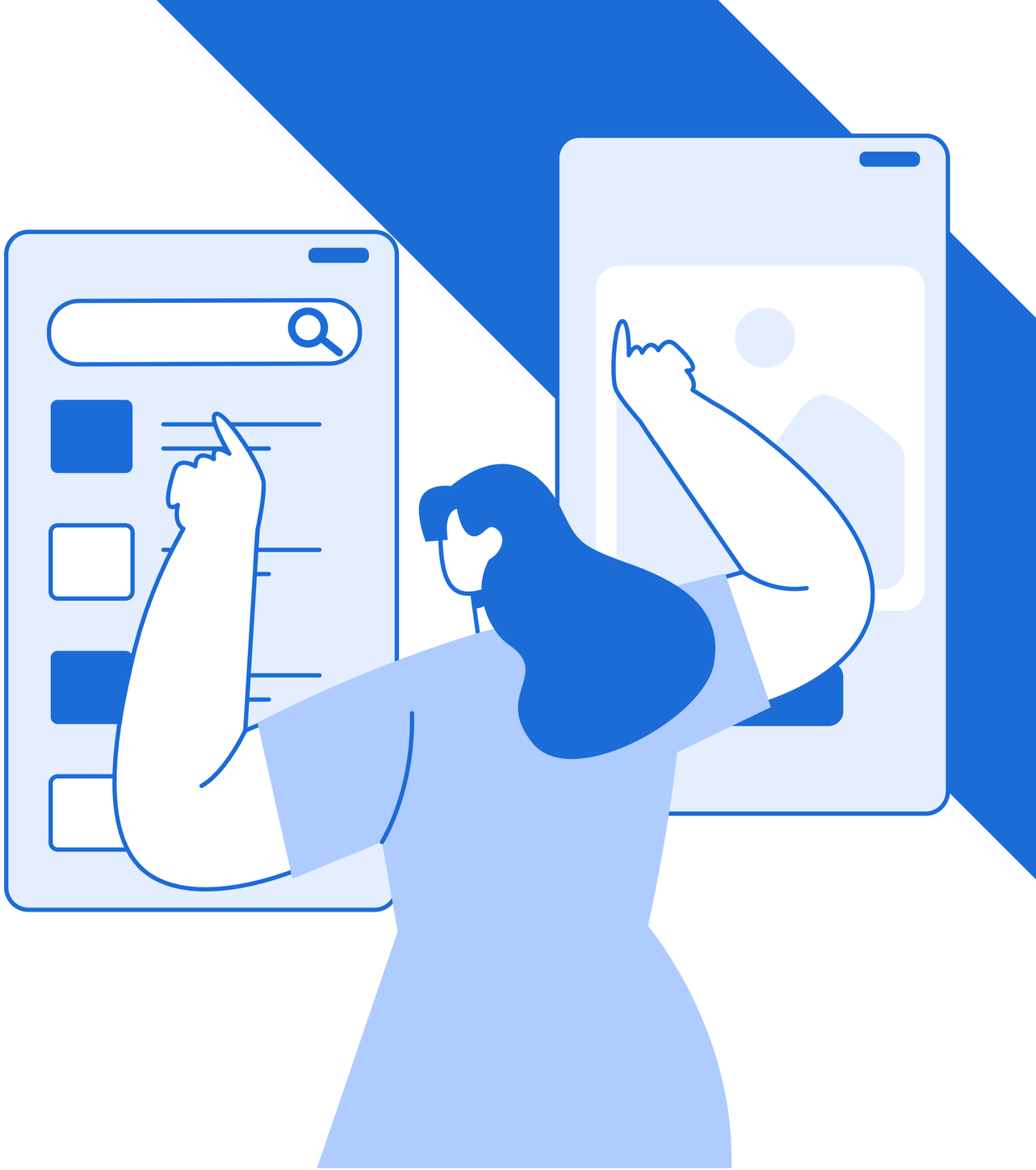
Hire Smart, Pay Right:

How a **Predictive Salary Model** Can Help Startup Companies Compete for **Top Analyst Talent**



DataBridge Analytics

- A private data analytics **consulting firm** based in Chicago, Michigan, that serves clients across the United States.

Who to hire?
What to pay?

- Only **1 in 5** companies formally tracks the quality of their hires (SHRM, 2025)
- **53%** of candidates considers higher compensation as primary decision driver when accepting an offer (Gartner, 2025)
- **54%** of job seekers have actively considered leaving their current role for better pay (BambooHR, 2025)

Meet Maria Cruz

- CEO of **NourishNation**
- A nonprofit company based in **Austin, Texas**, dedicated to combating child malnutrition across the United States. Closing the gap on hunger, one meal at a time.
- Company size: **28 employees**

Meet Maria Cruz

- Won a grant that required her to **hire new Data Analysts fast**
- **Roles needed:** 1 senior and 3 junior analysts (remote work)
- **Target skills:** SQL, Python, and Tableau

The Problem:

No Salary Benchmark

No Salary Benchmark

- She has **never recruited** from the open job market
- All previous hires were made through **referrals or personal networks**
- She has no HR manager, no recruiter, and **no formal salary structure**
- Compensation was always **negotiated** case by case

How much should she offer?

To determine exactly what to offer data analyst roles—without overpaying or losing talent to competitors

Dataset Preparation

Data Foundation Source:

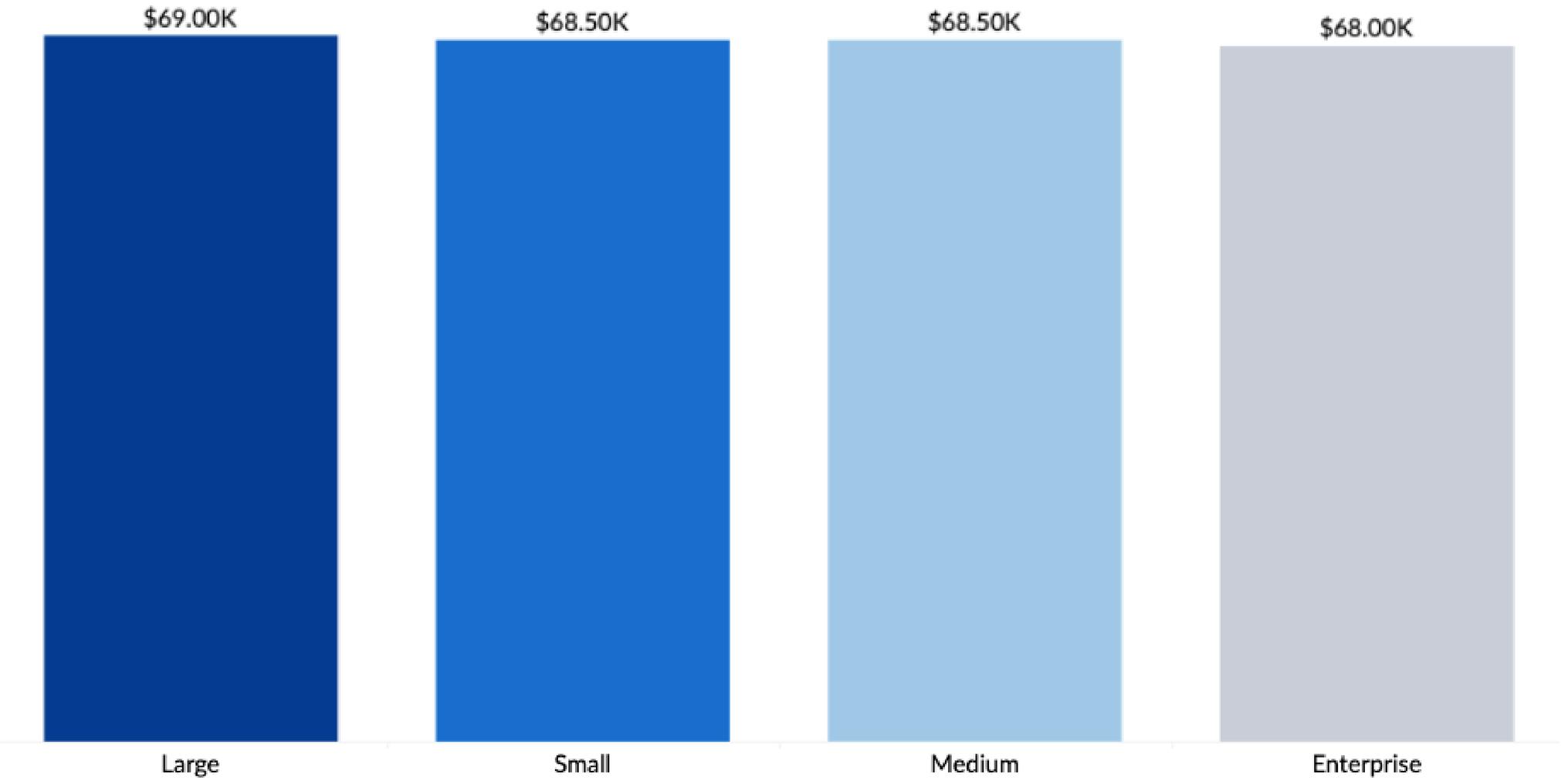
- **Source**: 1,671 Glassdoor Job Postings (Filtered specifically for "Data Analyst" roles).
- **Variables**: 17 Refined Features (Salary, Industry, Seniority, and Company Size).
- **Statistical Methods:** Descriptive Statistics to identify current market trends and Inferential Regression Modeling to predict competitive salary offers.

Dataset Preparation

Python ETL Pipeline (Pandas)

- **Salary Parsing:** Normalized text-based estimates and "K" abbreviations into clean numerical values (Min, Max, and Average)
- **Data Cleaning:** Used RegEx-based parsing to scan job descriptions, ensuring the dataset was strictly limited to roles matching Maria's technical requirements.
- **Standardization (Feature Engineering):** Categorized and standardized Seniority levels (Junior to Senior) and Company Size (Small to Enterprise).

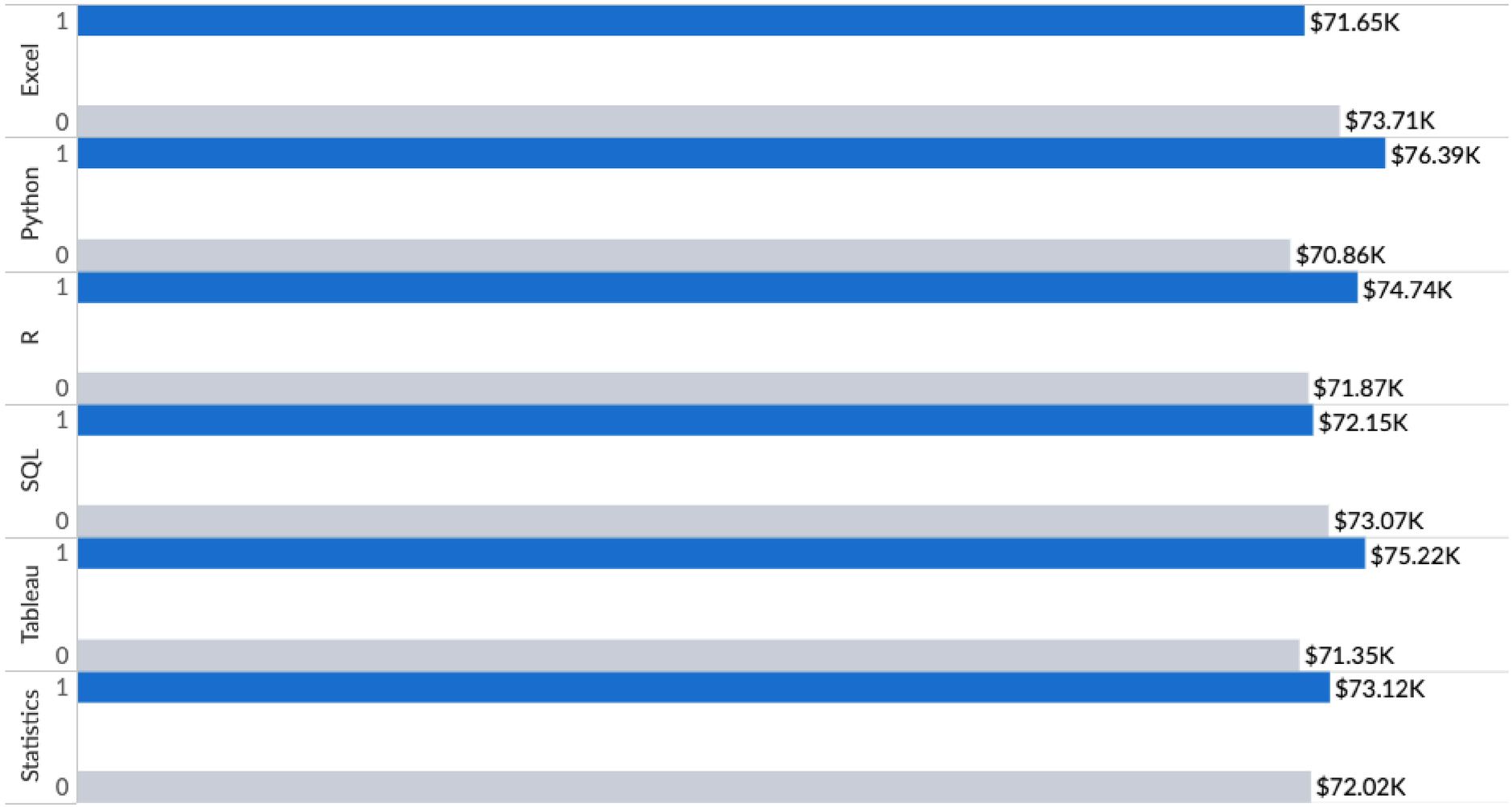
Salary offer remains standardized regardless of company size. ($68-$69K)



Sector is one of the key salary drivers. Databridge must align offers with target sector benchmark.

There is a significant difference in salary depending on job levels, requiring Databridge to have a clear salary tier system

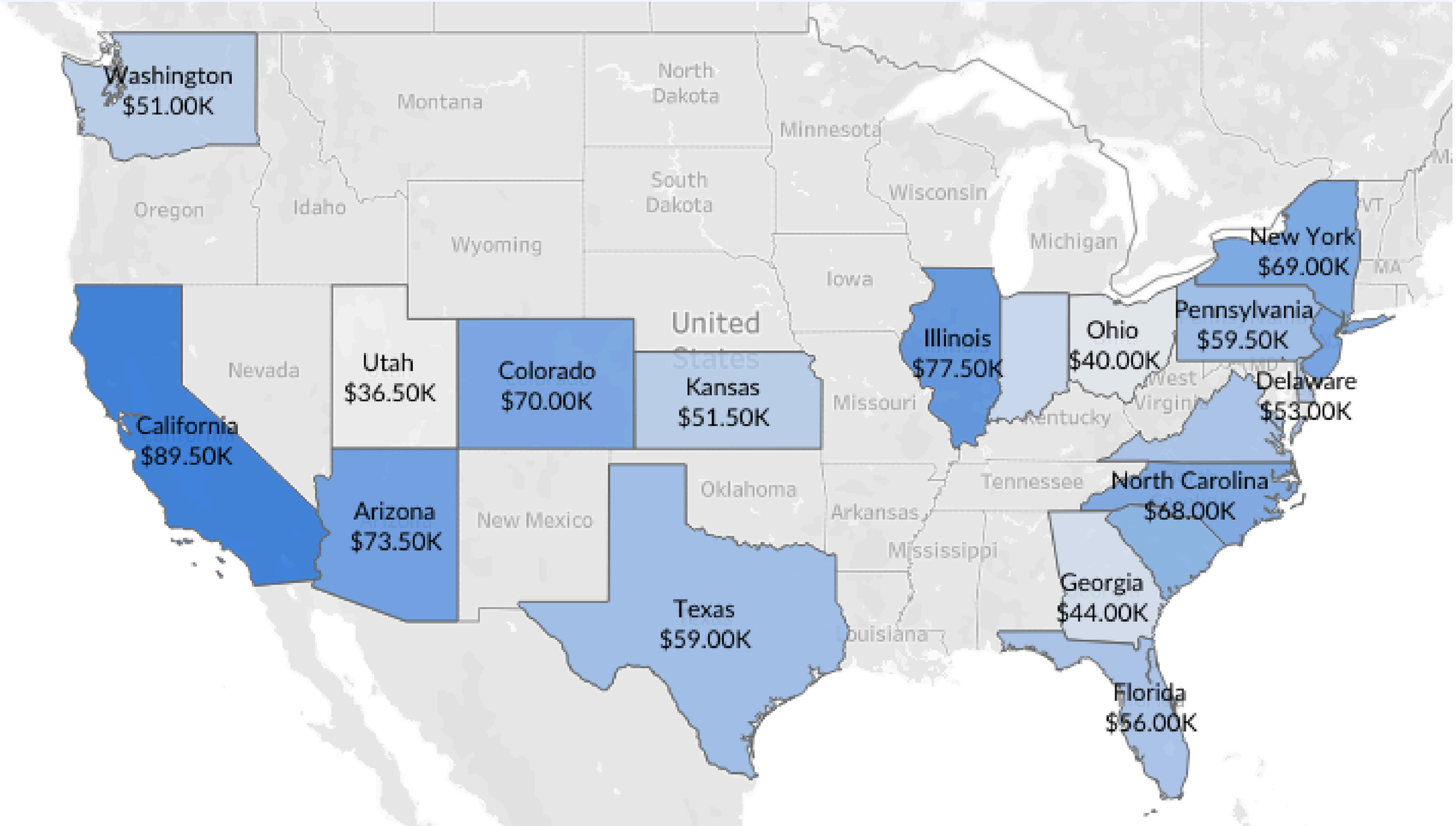
Jobs requiring certain skills lead to a higher salary (Python, Tableau, R)



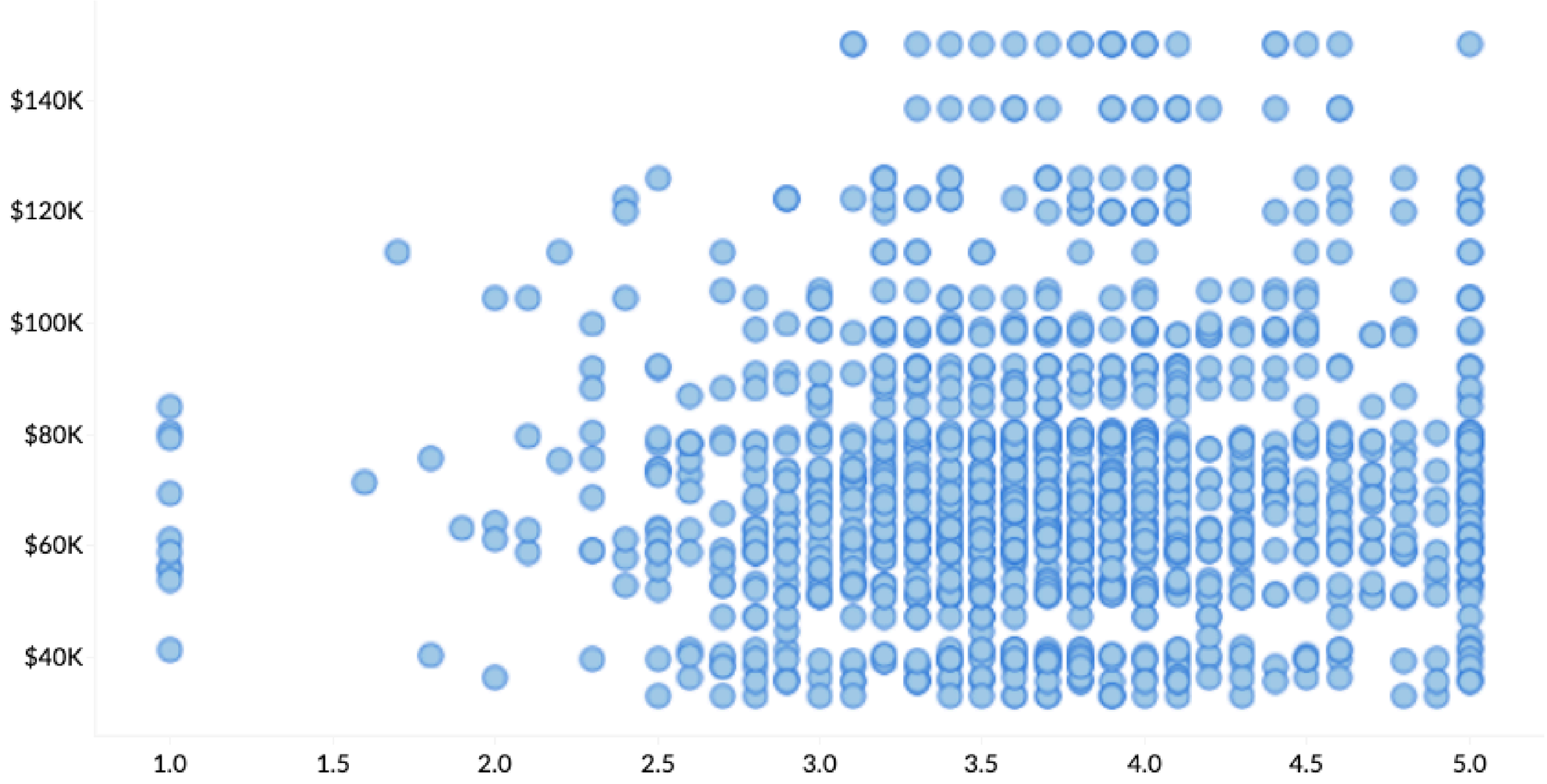
*1- requires skill; 0- does not require skill

Databridge's position is advantageous in terms of their ideal salary offer as a Private Practice / Firm. ($70.50K)

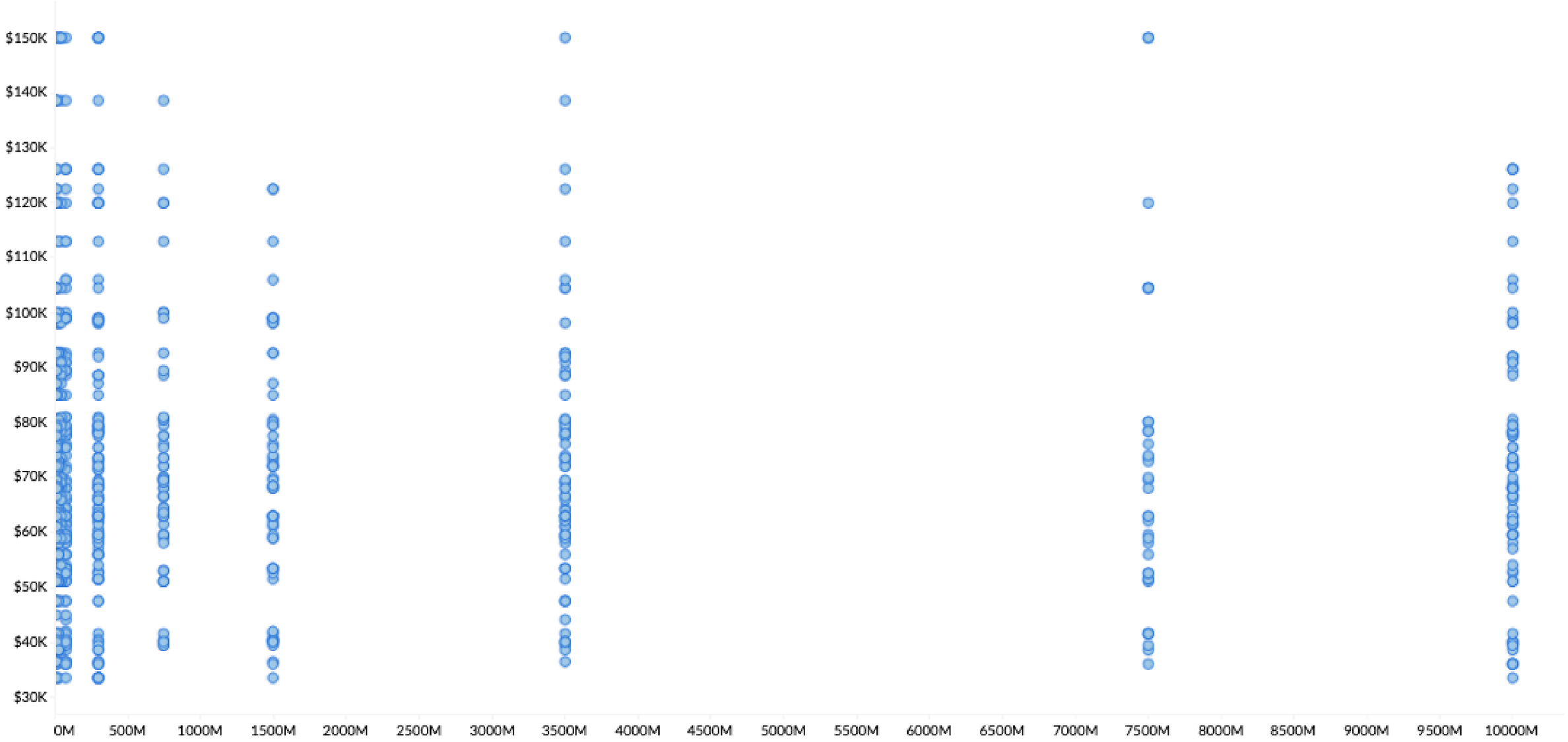
Location drives salary variation. Remote roles create an opportunity to optimize compensation strategy.

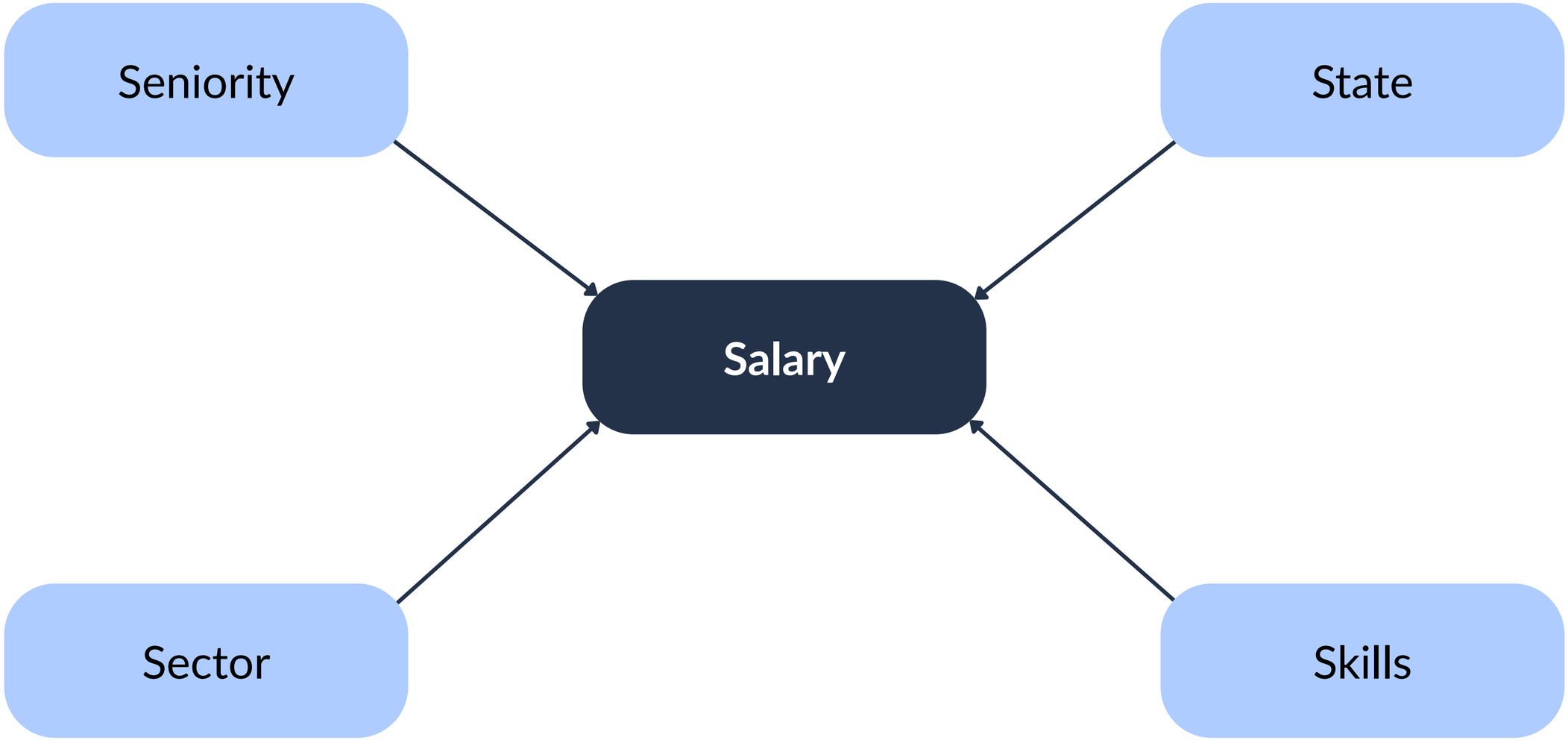


Company rating has a limited impact on salary



Higher Company Revenue ≠ Higher Salary Offer




Seniority
State
Salary
Sector
Skills

Regression Analysis

- Examine interdependencies among variables
- Create predictive model for the series.
- Enables the estimation of relationships between dependent and independent variables, providing a basis for forecasting.

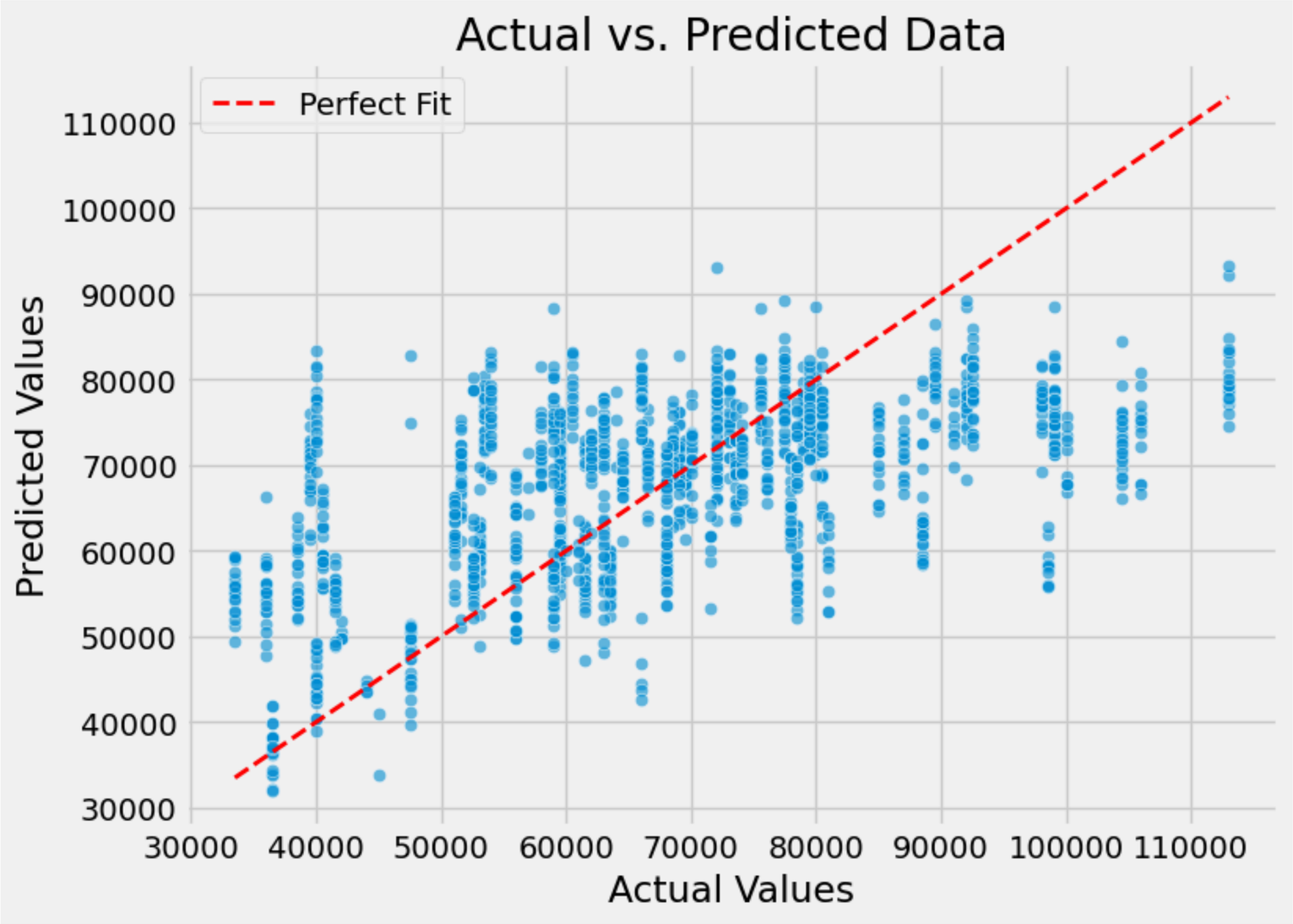
Regression Analysis

- Outliers were removed using the interquartile range (IQR) method
- Some sectors were grouped together to reduce multicollinearity

Regression Assumptions

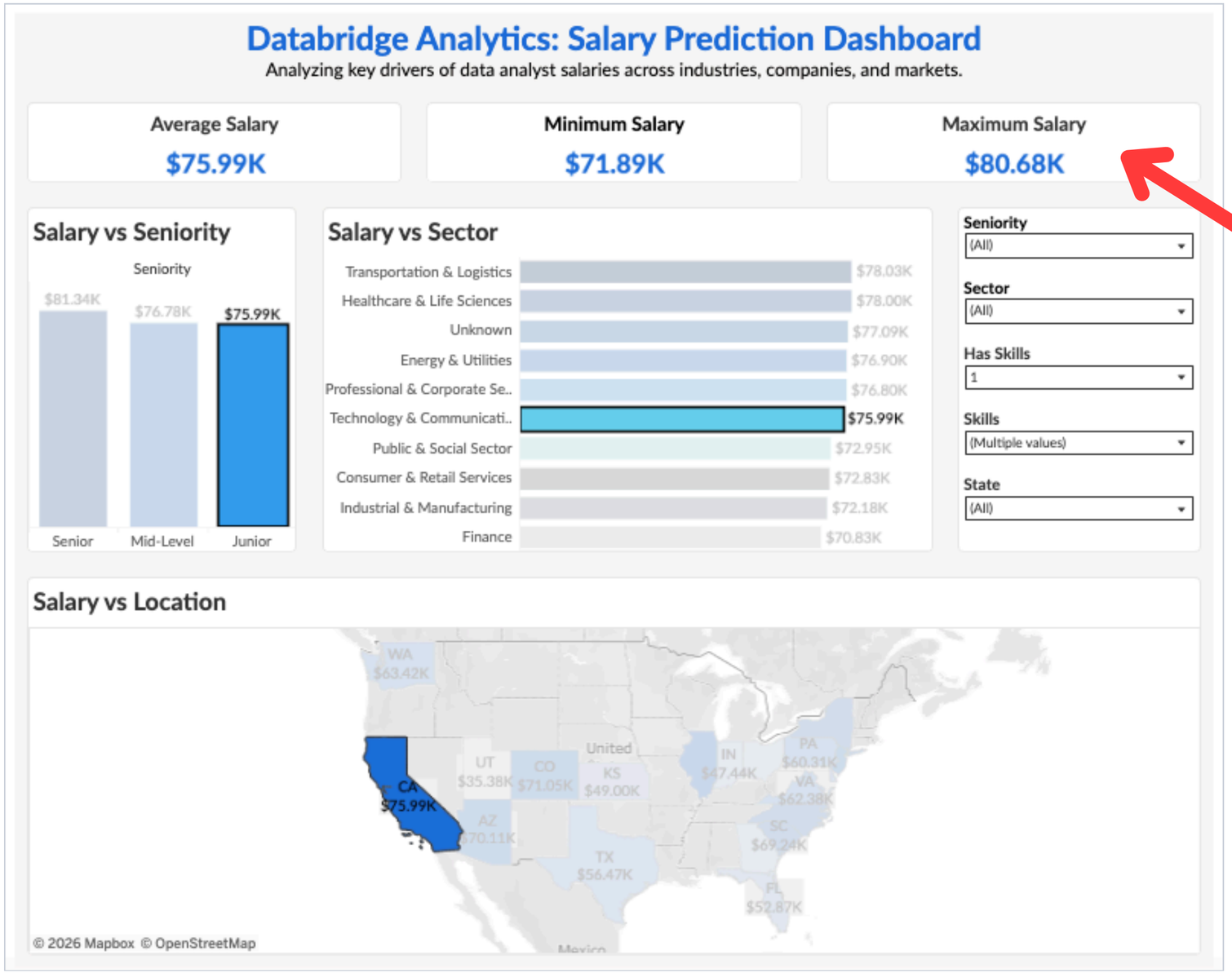
- ✓ Linearity
- ✗ Normality
  - remedied by enforcing the central limit theorem
- ✗ Homoscedasticity
  - remedied by using robust standard errors
- ✓ Autocorrelation
- ✓ Multicollinearity

The adjusted R2 suggests that 29.9% of the variability in the dataset can be explained by the model



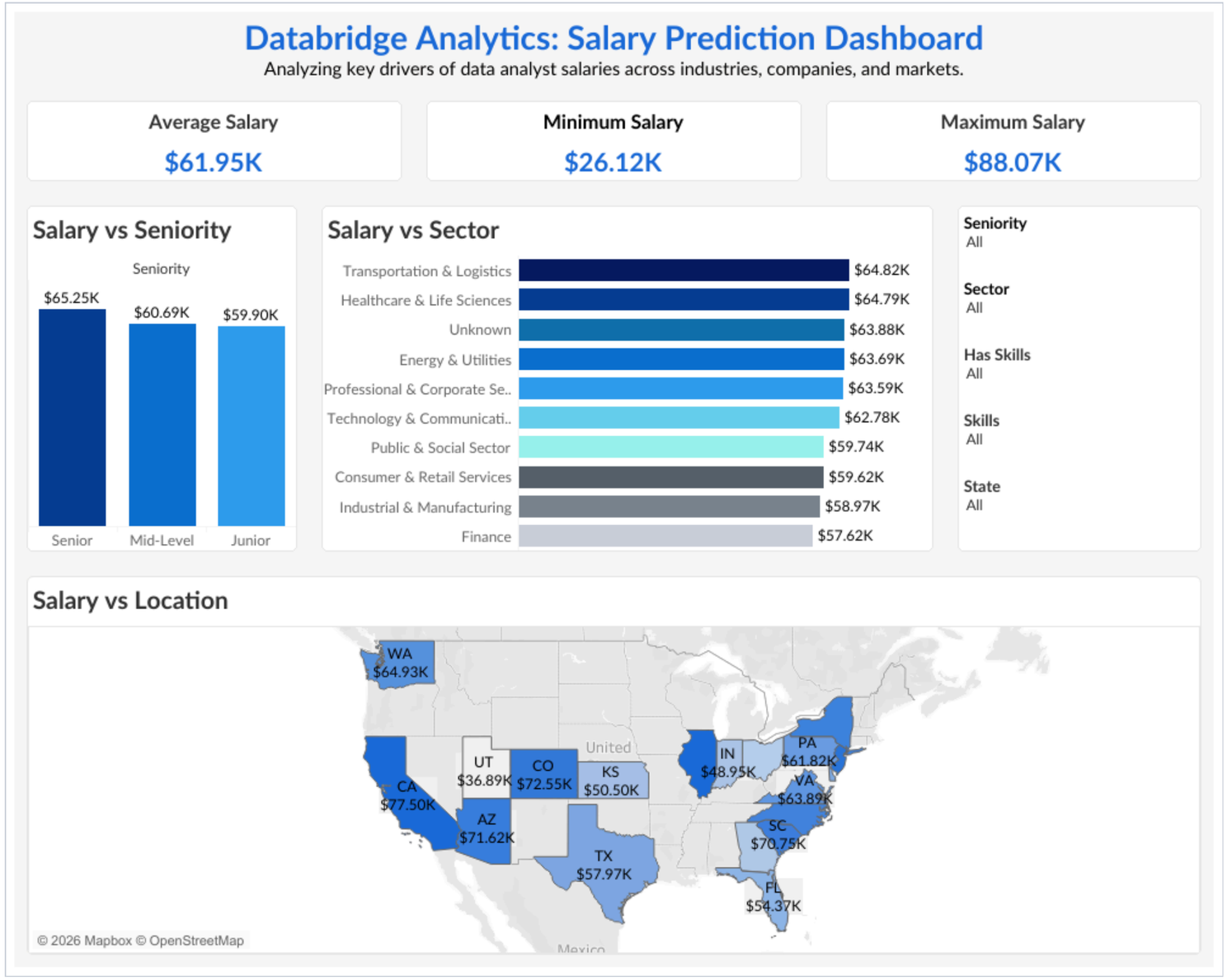
Company: Moveworks
Location: California
Sector: Technology
Role: Junior Data Analyst
Skills: SQL, Python, Tableau
Salary Offered: **$150K**

*Source: Glassdoor



- 46% higher than maximum salary
- Almost 2x higher than average salary
- Overspending on employee salary = waste of company budget

From guesswork, to data-driven salary decisions



You don’t need to outpay the market—you just need to understand it better.

Thank you

References

- Society for Human Resource Management. (2025). SHRM releases 2025 benchmarking reports: How does your organization compare? https://www.shrm.org/about/press-room/shrm-releases-2025-benchmarking-reports--how-does-your-organizat
- Gartner. (2025, June 16). Gartner HR research finds 44% of prospective candidates received multiple job offers in Q1 2025. https://www.gartner.com/en/newsroom/press-releases/2025-06-16-gartner-hr-research-finds-44-percent-of-prospective-candidates-received-multiple-job-offers-in-1q250
- BambooHR. (2025). 2025 compensation trends report. https://www.bamboohr.com/resources/guides/compensation-trends